



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

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Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

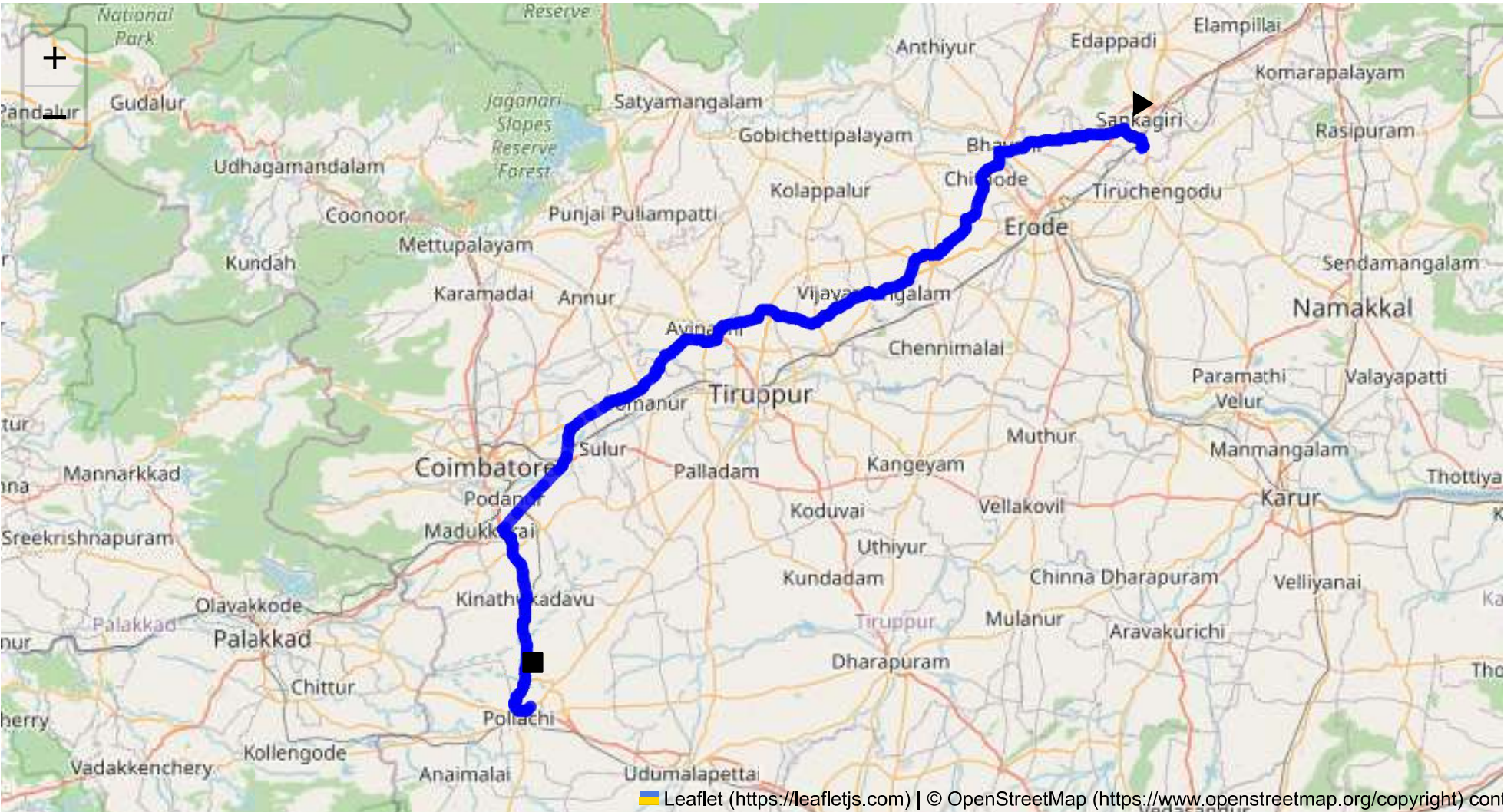
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 163.12 km
Estimated Duration: 3.0 hours
Adjusted Duration (Heavy Vehicle): 3.8 hours
Start: (11.4381, 77.8734)
End: (10.673099, 77.025467)



Risk Spots

	RiskSpotType	Coordinates	Speed Limit	Distance from Start
1	Turn	11.43968, 77.87345	15 KM/Hr	0.09 km
2	Turn	11.44024, 77.87528	15 KM/Hr	0.33 km
3	Turn	11.44898, 77.87410	15 KM/Hr	1.17 km

	RiskSpotType	Coordinates	Speed Limit	Distance from Start
4	Turn	11.45352, 77.85698	15 KM/Hr	3.26 km
5	Turn	11.45764, 77.85219	15 KM/Hr	3.89 km
6	Turn	11.46303, 77.84945	15 KM/Hr	4.61 km
7	Turn	11.44965, 77.77565	15 KM/Hr	12.86 km
8	Turn	11.44953, 77.77344	15 KM/Hr	13.12 km
9	Turn	11.43165, 77.67734	15 KM/Hr	24.15 km
10	Turn	11.39771, 77.65194	15 KM/Hr	29.21 km
11	Turn	11.34124, 77.63222	15 KM/Hr	36.47 km
12	Turn	11.33904, 77.63045	15 KM/Hr	36.92 km
13	Turn	11.20705, 77.37253	15 KM/Hr	72.18 km
14	Turn	11.18808, 77.28672	15 KM/Hr	83.17 km
15	Turn	11.17768, 77.28370	15 KM/Hr	84.03 km
16	Turn	11.17593, 77.28219	15 KM/Hr	84.62 km
17	Turn	11.17292, 77.26880	15 KM/Hr	86.10 km
18	Turn	11.17571, 77.24002	15 KM/Hr	89.30 km
19	Blind Spot	10.91785, 76.98617	10 KM/Hr	130.86 km
20	Turn	10.91510, 76.98975	15 KM/Hr	131.29 km
21	Turn	10.90412, 76.99831	15 KM/Hr	133.02 km
22	Turn	10.68391, 77.00591	15 KM/Hr	158.32 km
23	Turn	10.67921, 77.00392	15 KM/Hr	158.86 km
24	Turn	10.66766, 77.00751	15 KM/Hr	160.11 km
0	Roundabout	10.66790, 77.01218	15 KM/Hr	160.84 km
25	Turn	10.66875, 77.01287	15 KM/Hr	160.93 km
26	Turn	10.66797, 77.01370	15 KM/Hr	161.04 km
27	Turn	10.66760, 77.01825	15 KM/Hr	161.58 km

Emergency Services

	type	name	coordinates	side_of_road
0	hospital	Dhanvanthri Hospital	11.4496712, 77.7593772	Opposite Side
1	hospital	J.K.K. Trust Hospital	11.4445841, 77.7307962	Driver's Side
2	hospital	Shri Sathyanarayana Hospital	11.4291297, 77.6913408	Driver's Side

	type	name	coordinates	side_of_road
4	hospital	Thanish Siddha Hospital	11.430003, 77.674964	Opposite Side
5	clinic	Harshitha Clinic	11.4313207, 77.674718	Opposite Side
6	hospital	Sri Kaalangi Siddhar Mooligai Vaithiya Nilayam	11.432369, 77.674894	Opposite Side
7	clinic	G.K Clinic	11.4297244, 77.6749715	Opposite Side
10	clinic	Erode Cancer Centre	11.3732, 77.649152	Opposite Side
20	hospital	Irt Hospital	11.2803603, 77.5644118	Driver's Side
24	hospital	Gen Siddha Hospital	11.2375033, 77.5059797	Opposite Side
25	hospital	Vijayamangalm Government Hospital	11.2382827, 77.501687	Opposite Side
26	hospital	Dr. N Viswanathan Hospital	11.2412783, 77.5005502	Opposite Side
28	clinic	P.M. Clinic	11.2281078, 77.4647296	Opposite Side
30	hospital	Sri Renu Hospital	11.1990225, 77.4219937	Driver's Side
34	police	Karumathampatti Police Station	11.1093464, 77.1802095	Driver's Side
44	hospital	Saraswathi Hospital	11.0069868, 77.071368	Driver's Side
52	hospital	Ambal Malumichampatty, Coimbatore	10.90265, 76.997586	Opposite Side
53	hospital	Sri Venkateswara Kinnathukadavu	10.8241085, 77.0191884	Opposite Side
54	hospital	Government Hospital, Kinathukadavu	10.82107, 77.017406	Opposite Side
56	hospital	The Pollachi Cardiac Center	10.6742907, 77.0081305	Driver's Side
57	hospital	K.G.M. Hospital Paavhai Women's Cente, Pollachi	10.67023, 77.007418	Driver's Side
58	hospital	Arathana Hospital	10.6682257, 77.0070174	Driver's Side
59	clinic	A.S.D. Nursing Home	10.6665361, 77.0084792	Opposite Side

	type	name	coordinates	side_of_road
60	clinic	Andavar Clinic and Nursing Home	10.6679726, 77.0081641	Driver's Side
61	clinic	Neuro Poly Clinic, Pollachi	10.6662216, 77.0073506	Opposite Side
62	hospital	Thirumalai Hospital	10.6650579, 77.005872	Opposite Side
64	clinic	Syndhiya Ortho Clinic	10.6663252, 77.0158674	Opposite Side

Rest Stops

	type	name	coordinates	side_of_road
3	fuel	Reliance	11.4325168, 77.6912865	Opposite Side
8	fuel	Indian Oil	11.4321177, 77.6765153	Opposite Side
9	fuel	KR Fuels - Auto LPG	11.4127737, 77.6747462	Driver's Side
11	hotel	Athithi cafe	11.3756993, 77.6516401	Driver's Side
12	restaurant	Athithi Cafe	11.3756705, 77.6516565	Driver's Side
13	restaurant	Paradise Biriyani	11.3291744, 77.6297833	Driver's Side
14	parking	NHAI parking	11.3030168, 77.6009269	Opposite Side
15	hotel	Hotel Annachi	11.3020259, 77.600017	Opposite Side
16	hotel	Adayar Ananda Bhavan(a2b)	11.3018451, 77.599909	Opposite Side
17	restaurant	Ambur Star Biriyani	11.3017372, 77.5999546	Opposite Side
18	restaurant	Arukaniyammal Mess	11.2982771, 77.5967424	Opposite Side
19	fuel	HP	11.2909675, 77.5815262	Driver's Side
21	fuel	Reliance	11.2575595, 77.5511245	Driver's Side
22	fuel	Bharat Petroleum	11.255444, 77.5457729	Opposite Side
23	fuel	Auto LPG	11.2523222, 77.5400243	Opposite Side
27	fuel	HP Gas Station	11.239607, 77.4949754	Driver's Side
29	fuel	Indian Oil	11.2126458, 77.4417156	Opposite Side
31	restaurant	Shri Sai Krishna	11.1993912, 77.3207179	Driver's Side
32	fuel	HP bunk	11.1946674, 77.2943298	Driver's Side
33	fuel	Bharat Petroleum	11.1673995, 77.2320569	Driver's Side
35	fuel	Bharat Petroleum	11.1050354, 77.1740448	Driver's Side
36	restaurant	Sri Divya Bakes	11.0917541, 77.1398745	Opposite Side

	type	name	coordinates	side_of_road
37	fuel	Bharat Petroleum	11.0740038, 77.10848	Opposite Side
38	cafe	Sri Divya Bakes	11.0722861, 77.1072797	Driver's Side
39	cafe	Sri Divya Bakes	11.0724244, 77.1074386	Driver's Side
40	restaurant	MDS Bakes	11.0567194, 77.0846726	Driver's Side
41	restaurant	Haryana Dhaba	11.046157, 77.0784461	Driver's Side
42	restaurant	KVK Bakes & Restaurant	11.0457521, 77.0778414	Opposite Side
43	fuel	Hindustan Petroleum	11.0089785, 77.070763	Opposite Side
45	restaurant	Kovai Gate	11.00112, 77.0661808	Opposite Side
46	cafe	Kaapi 2.0	11.0010379, 77.0661236	Opposite Side
47	restaurant	MDS Bakes	11.0020155, 77.067235	Driver's Side
48	restaurant	Joji Restaurant	10.9976871, 77.0628899	Opposite Side
49	fuel	Arumuga Konar Agencies	10.9740857, 77.0399491	Opposite Side
50	restaurant	Bay Grill Restaurant and Cafe	10.9183046, 76.9862353	Driver's Side
51	restaurant	Saravana Bhavan	10.9179855, 76.9866804	Driver's Side
55	fuel	Indian Oil	10.8064491, 77.0174353	Driver's Side
63	restaurant	The Slaves	10.667635, 77.0092111	Opposite Side

Welcome to the Journey Risk Management Study

Route Overview: The route from Sangagiri to Pollachi, Tamil Nadu, predominantly involves travel along major highways and state roads, spanning approximately 163.12 kilometers and taking about 3 hours under typical conditions. The journey primarily navigates through rural and semi-urban settings.

1. Overview of the Route Map: The path typically involves traveling southward from Sangagiri, potentially via State Highway (SH 86) to NH 544, and continuing south through Erode, Gobichettipalayam, and Coimbatore edges before reaching Pollachi. The route includes several arterial roads, connecting through key junctions that see relatively higher traffic, particularly near urban centers.

2. Weather Conditions: Tamil Nadu generally experiences tropical weather, with temperatures ranging from mild to high. Monsoon season (June-September) may pose challenges due to heavy rains, leading to slippery roads and reduced visibility. In the summer months (March-May), high temperatures could affect vehicle performance.

3. Traffic Patterns: Traffic congestion is typically observed during morning (8-10 AM) and evening (6-8 PM) hours, especially near urban areas like Erode and outskirts of Coimbatore. Major intersections, toll plazas, and city entry/exit points experience bottlenecks, particularly around industrial zones.

4. Road Quality and Infrastructure: The route primarily utilizes national and state highways, maintained reasonably well for heavy vehicular movement. However, sections might have uneven surfaces, especially in more rural areas. Under-construction or repaired sections can pose temporary hazards, requiring caution.

- 5. Alternative Routes for Emergencies:** In case of roadblocks or heavy congestion, alternative routes bypassing busy towns could include smaller state roads running parallel south of NH 544, albeit requiring slower speeds and more caution given potentially poorer road conditions.
- 6. Local Regulations for Hazardous Material Transport:** Transport of hazardous materials in Tamil Nadu is regulated stringently, requiring compliance with labeling, handling, and routing protocols. Restrictions on timing and routing through densely populated or environmentally sensitive areas may apply.
- 7. Historical Incidents:** There have been sporadic reports of accidents involving heavy vehicles on these routes, typically due to driver fatigue, technical failure, or adverse weather. Transporting hazardous materials increases risks, necessitating adherence to strict safety regulations.
- 8. Environmental Considerations:** The route crosses agricultural lands and occasionally skirts wildlife-oriented regions. Minimizing emissions and avoiding spillage is critical to protect these areas. Compliance with noise and air pollution standards is essential.
- 9. Communication Coverage:** Most of the route is covered adequately by mobile networks, except potential short dead zones in more remote rural stretches. It is advisable to have alternate communication tools or GPS devices that function offline if necessary.
- 10. Emergency Response Times:** In urban or semi-urban stretches near towns like Erode and Coimbatore, emergency services are relatively quick (within 30 minutes). In rural sections, response times may extend up to an hour. Carrying a basic emergency kit and having contact information for local services is advisable.

Overall Summary of Risk Assessment: Due to generally well-maintained infrastructure and comprehensive regulation, the route poses moderate risk under favorable weather. Key hazards include monsoon-induced slippery conditions, congestion in urban areas, and the inherent risk of transporting hazardous materials. Regular review of route conditions and contingency planning for emergencies can help in mitigating risks effectively. It's crucial to stay updated on weather forecasts and traffic advisories.

Route Photos of Risky Spots



Risk Type: Turn

Speed Limit: 15 KM/Hr

Distance from Start: 0.33 km

Coordinates: 11.44024, 77.87528



Risk Type: Turn

Speed Limit: 15 KM/Hr

Distance from Start: 1.17 km

Coordinates: 11.44898, 77.87410



Risk Type: Turn

Speed Limit: 15 KM/Hr

Distance from Start: 3.26 km

Coordinates: 11.45352, 77.85698



Risk Type: Turn

Speed Limit: 15 KM/Hr

Distance from Start: 3.89 km

Coordinates: 11.45764, 77.85219



Risk Type: Turn

Speed Limit: 15 KM/Hr

Distance from Start: 4.61 km

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Speed Limit: 15 KM/Hr

Distance from Start: 12.86 km

Coordinates: 11.44965, 77.77565



Risk Type: Turn

Speed Limit: 15 KM/Hr

Distance from Start: 13.12 km

Coordinates: 11.44953, 77.77344



Risk Type: Turn
Speed Limit: 15 KM/Hr
Distance from Start: 24.15 km
Coordinates: 11.43165, 77.67734



Risk Type: Turn
Speed Limit: 15 KM/Hr
Distance from Start: 29.21 km
Coordinates: 11.39771, 77.65194



Risk Type: Turn
Speed Limit: 15 KM/Hr
Distance from Start: 36.47 km
Coordinates: 11.34124, 77.63222



Risk Type: Turn
Speed Limit: 15 KM/Hr
Distance from Start: 36.92 km
Coordinates: 11.33904, 77.63045



Risk Type: Turn
Speed Limit: 15 KM/Hr
Distance from Start: 72.18 km
Coordinates: 11.20705, 77.37253



Risk Type: Turn
Speed Limit: 15 KM/Hr
Distance from Start: 83.17 km
Coordinates: 11.18808, 77.28672



Risk Type: Turn
Speed Limit: 15 KM/Hr
Distance from Start: 84.03 km
Coordinates: 11.17768, 77.28370



Risk Type: Turn
Speed Limit: 15 KM/Hr
Distance from Start: 84.62 km
Coordinates: 11.17593, 77.28219



Risk Type: Turn
Speed Limit: 15 KM/Hr
Distance from Start: 86.10 km
Coordinates: 11.17292, 77.26880



Risk Type: Turn
Speed Limit: 15 KM/Hr
Distance from Start: 89.30 km
Coordinates: 11.17571, 77.24002



Risk Type: Blind Spot

Speed Limit: 10 KM/Hr

Distance from Start: 130.86 km

Coordinates: 10.91785, 76.98617



Risk Type: Turn

Speed Limit: 15 KM/Hr

Distance from Start: 131.29 km

Coordinates: 10.91510, 76.98975



Risk Type: Turn

Speed Limit: 15 KM/Hr

Distance from Start: 133.02 km

Coordinates: 10.90412, 76.99831



Risk Type: Turn

Speed Limit: 15 KM/Hr

Distance from Start: 158.32 km

Coordinates: 10.68391, 77.00591



Risk Type: Turn

Speed Limit: 15 KM/Hr

Distance from Start: 158.86 km

Coordinates: 10.67921, 77.00392



Risk Type: Turn
Speed Limit: 15 KM/Hr
Distance from Start: 160.11 km
Coordinates: 10.66766, 77.00751



Risk Type: Roundabout
Speed Limit: 15 KM/Hr
Distance from Start: 160.84 km
Coordinates: 10.66790, 77.01218



Risk Type: Turn
Speed Limit: 15 KM/Hr
Distance from Start: 160.93 km
Coordinates: 10.66875, 77.01287



Risk Type: Turn
Speed Limit: 15 KM/Hr
Distance from Start: 161.04 km
Coordinates: 10.66797, 77.01370



Risk Type: Turn
Speed Limit: 15 KM/Hr
Distance from Start: 161.58 km
Coordinates: 10.66760, 77.01825

[Download Interactive Map](#)

